

Assignment-3.1

“Tools for troubleshooting”

Task 1 List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case

1) Testing a Laptop Power Adapter

Application: You switch the multimeter to DC voltage mode and touch the probes to the charger plug pin and barrel.

How it helps: It checks if the power brick sends out the correct voltage (like 19.5V), showing whether the charger is broken or if the laptop has an internal power port issue.

2) Finding Short Circuits on a Motherboard

Application: You switch the multimeter to continuity mode (the beep setting) and touch small board parts and ground points.

How it helps: It makes a beep sound when electricity flows straight to ground, helping you spot shorted parts that stop the computer from turning on.

3) Checking Smartphone Battery Health

Application: You touch the multimeter probes directly to the small battery connector contacts.

How it helps: It checks if the battery holds enough power (around 3.7V to 4.2V) to power up the phone, so you know if you need a new battery.

Task 2 Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Focus on the codes shown and how they relate to hardware issues.

- first plug the diagnostic POST card into the PCI or PCIe slot of motherboard before turning on the computer. POST help us to diagnose hardware problems, by showing unique codes.
- It shows two-digit code numbers on a small light display. few codes and their related issues are:
 1. 00Ac → ALL OK.
 2. d454 → with NO display means → there is RAM issue → u can replace it with new ram and problem can be solved.
 3. 0000 → no display → CPU not detected → clean (if cpu is already there)
 4. E098 → no display → RAM not detected → clean slots and check locks or if it is assembled correctly. (if ram is placed already).

TASK 3 Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Blur or crop out any personal information before submitting.</em

CrystalDiskInfo 9.9.2 x64

File Edit Function Theme Disk Help Language

Good
41 °C
C: D:

Phison ESO512GMLCT-E9C : 512.1 GB

Good
89 %

Temperature
41 °C

Firmware		Total Host Reads	12419 GB
Serial Number		Total Host Writes	6959 GB
Interface	NVM Express	Rotation Rate	---- (SSD)
Transfer Mode	PCIe 4.0 x4 PCIe 4.0 x4	Power On Count	640 count
Drive Letter	C: D:	Power On Hours	631 hours
Standard	NVM Express 1.4		
Features	S.M.A.R.T., TRIM, VolatileWriteCache		

ID	Attribute Name	Raw Values
01	Critical Warning	00000000000000
02	Composite Temperature	0000000000013A
03	Available Spare	00000000000064
04	Available Spare Threshold	00000000000005
05	Percentage Used	0000000000000B
06	Data Units Read	000000018D6989
07	Data Units Written	00000000DEB377
08	Host Read Commands	0000000100DB962
09	Host Write Commands	0000000B4D1C07
0A	Controller Busy Time	00000000000033A
0B	Power Cycles	000000000000280
0C	Power On Hours	000000000000277
0D	Unsafe Shutdowns	00000000000000A
0E	Media and Data Integrity Errors	000000000000000
0F	Number of Error Information Log Entries	000000000000026
10	Warning Composite Temperature Time	000000000000000
11	Critical Composite Temperature Time	000000000000000
1A	Thermal Management Temperature 1 Transition Count	000000000000000

TASK 4 Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

TOOL	What it checks	Example problem detected
Multimeter	Raw electricity values like AC/DC voltage, electrical current, resistance, and wire connections.	A power supply unit giving low voltage (like 10V instead of 12V), or a tiny shorted part on the motherboard pulling down power.
POST Card	Early hardware setup tests run by the BIOS chip before the monitor screen turns on.	A computer failing to boot because of bad RAM sticks (code C1) or a corrupted BIOS chip.
Software Diagnostics CrystalDiskInfo	System sensor details inside Windows, including live heat numbers, fan speeds, voltage stability, and drive health indicators.	A processor getting too hot because of old thermal paste, or bad sectors on a hard drive warning that it might crash soon.